

On the bottom

THE BOTTOM OF THE DEEP OCEAN is not an easy place to live. There is little food and it is dark and cold. Much of the seabed is covered with soft clays or mudlike oozes made of skeletons of tiny sea animals and plants. The ooze on the vast open plains of the abyss can reach several hundred yards thick. Animals walking along the bottom have long legs to keep from stirring it up. Some grow anchored to the seabed and have long stems to keep their feeding structures clear of the ooze. Food particles can be filtered out of the water, for example, by the feathery arms in sea lilies or through the many pores in sponges. Some animals, such as sea cucumbers, feed on the seabed and manage to extract enough goodness from food particles within the ooze. Food particles are the remains of dead animals (and their droppings) and plants that have sunk down from above. Occasionally, a larger carcass reaches the bottom uneaten, providing a real bonanza for the mobile bottom-dwellers that home in on it from all around. Because food is scarce and temperatures so low, most animals living in the deep ocean take a long time to grow.



Underwater cables were laid across the Atlantic Ocean to relay telegraphic messages, c. 1870

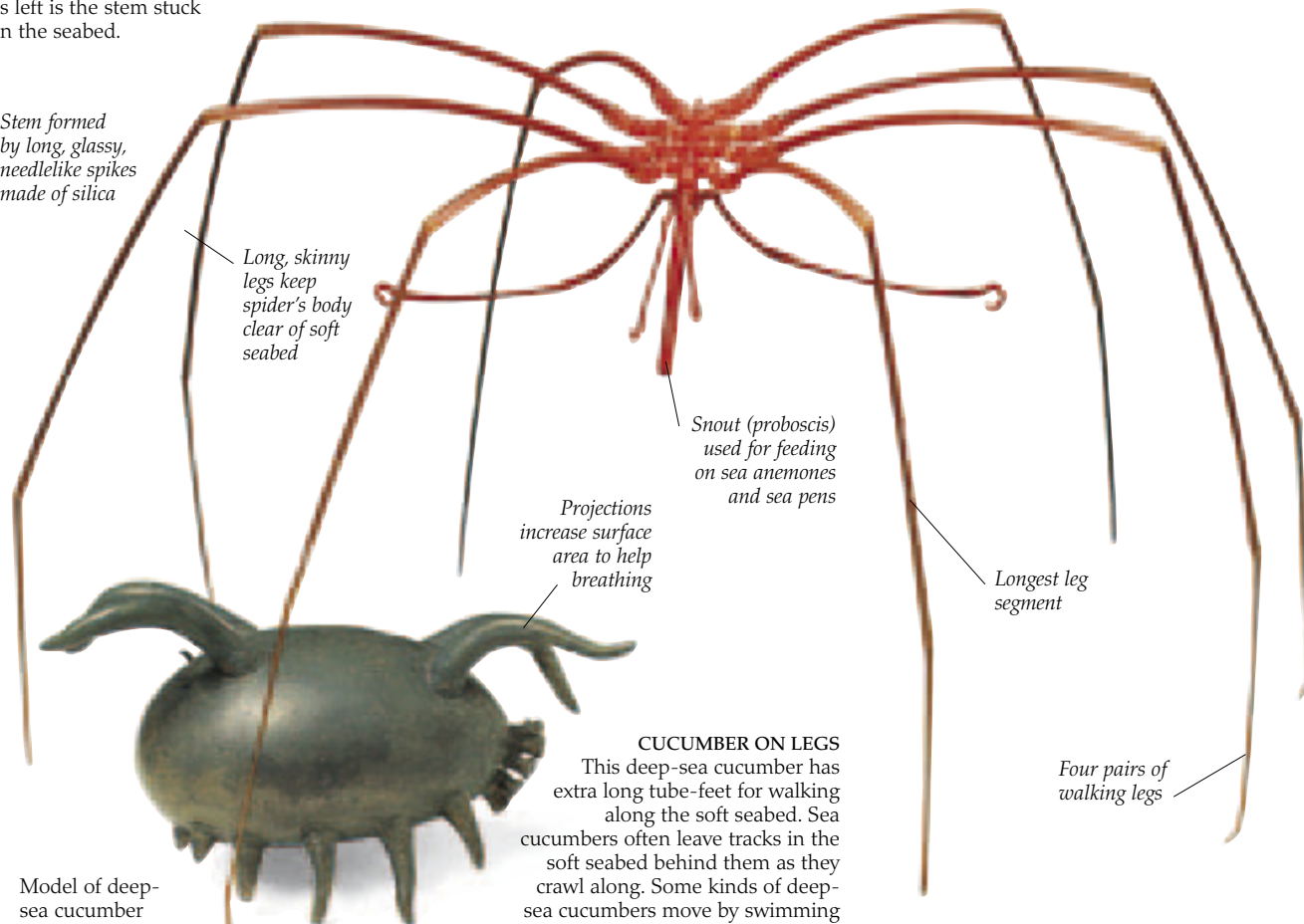
GLASSY STRANDS

This sponge grows anchored to the soft seabed by its stem of glass strands and sea anemones often grow on their stems. When a glass-rope sponge dies, the cup-shaped part disappears and all that is left is the stem stuck in the seabed.

Stem formed by long, glassy, needlelike spikes made of silica

NOT A TRUE SPIDER

Looking like land spiders, sea spiders belong to a group called pycnogonids. Some deep-sea spiders have a leg span of 2 ft (60 cm) across, and can stride along without stirring up clouds of particles. They can also swim, by launching off the seabed, bringing their legs upward, then sinking down again.



Long, skinny legs keep spider's body clear of soft seabed

Snout (proboscis) used for feeding on sea anemones and sea pens

Projections increase surface area to help breathing

Longest leg segment

Four pairs of walking legs

CUCUMBER ON LEGS

This deep-sea cucumber has extra long tube-feet for walking along the soft seabed. Sea cucumbers often leave tracks in the soft seabed behind them as they crawl along. Some kinds of deep-sea cucumbers move by swimming above the bottom.

Model of deep-sea cucumber (Scotoplanes)